

Software Requirements Specification for Sahayatri (v1.0 Approved)

सहायात्री — Verified Elder-Care & Remote Family Monitoring Platform

Revision History

Name	Date	Reason For Changes	Version
Sahayatri Core Architecture Team	July 2026	Initial baseline engineering spec for production design	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional, non-functional, and architectural requirements for **Sahayatri**, a three-sided verified elder-care and remote family monitoring platform connecting Nepali diaspora workers abroad with vetted, background-checked caregivers who physically visit and monitor elderly parents (care recipients) residing in Nepal.

The document covers the full release-candidate (RC-1.0) scope of three client applications and one backend platform:

- **Sahayatri Family** (React Native, iOS/Android) — used by diaspora payers to browse caregivers, subscribe to care plans, view proof-of-visit logs, receive SOS alerts, and manage billing in foreign currency.
- **Sahayatri Care** (React Native, iOS/Android) — used by caregivers/nurses to accept visit assignments, check in/out via GPS, upload proof-of-visit photographs, log vitals, and trigger emergency escalation.
- **Sahayatri Elder Companion** (React Native, Android-first, simplified UI, optional physical SOS pendant/hardware button pairing) — a lightweight app or companion device interface used directly by the elderly care recipient, designed for low digital literacy and low-bandwidth rural connectivity.
- **Sahayatri Core Platform** — the Node.js/Express backend, Socket.io real-time cluster, PostgreSQL/PostGIS data layer, background job workers, and third-party integration gateways (payments, communications, mapping, media storage) that all three clients depend on.

This SRS is scoped strictly to the engineering and business-analysis requirements needed to design, build, test, and operate RC-1.0. It excludes marketing collateral, caregiver recruitment HR policy outside the platform's verification workflow, and any hardware

manufacturing specification for the optional SOS pendant beyond its Bluetooth Low Energy (BLE) communication contract with the Elder Companion app.

1.2 Document Conventions

Requirement Tagging. Every atomic, testable requirement is tagged with a unique identifier following the pattern `REQ-<DOMAIN>-<NNN>`, where `<DOMAIN>` is one of:

Domain Code	Meaning
<code>FUNC</code>	Functional/business-logic requirement
<code>SEC</code>	Security requirement
<code>PERF</code>	Performance requirement
<code>DATA</code>	Data model / persistence requirement
<code>INT</code>	Third-party integration requirement
<code>UI</code>	User interface requirement
<code>OFFLINE</code>	Offline-first / sync requirement
<code>COMP</code>	Regulatory/compliance requirement

Requirements are numbered sequentially within each System Feature section (e.g., `REQ-FUNC-101` through `REQ-FUNC-118` under Feature 4.1) and are never renumbered or reused once assigned in a released version, to preserve traceability across sprints and QA test-case mapping.

Priority Convention. Each requirement carries a priority marker:

- **[P0 — Blocking]:** RC-1.0 cannot ship without this. A missing P0 fails the release gate.
- **[P1 — High]:** Required for the RC-1.0 target markets (GCC, Malaysia, Korea corridors) but may slip one sprint with Product sign-off.
- **[P2 — Deferred]:** Targeted for v1.1+, documented here for architectural forward-compatibility (e.g., schema fields reserved).

Formatting Standards. Requirement statements use the RFC 2119 keywords **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, and **MAY** with their standard normative meaning.

Database entities are written in `PascalCase`. API endpoints are written in `lower-kebab-case` prefixed with version, e.g., `/v1/visits/{visitId}/check-in`.

1.3 Intended Audience and Reading Suggestions

Audience	Recommended Reading Path
Backend/Mobile Developers	Sections 2.1, 2.4, 2.5 → Section 3 (all) → Section 4 (all, with focus on 4.x.3 REQ tables) → Section 5.1, 5.3
QA/Test Engineers	Section 4 (all Stimulus/Response tables) → Section 5.1, 5.2, 5.5 → Appendix B (Use Case Mapping)
DevOps/SRE	Sections 2.4, 2.5 → Section 3.4 → Section 5.1, 5.2, 5.4 → Appendix B (DFDs)
System/Database Administrators	Section 2.5 → Appendix B (ERD) → Section 6 (Data Retention) → Section 5.3 (RBAC matrix)
Academic Reviewers / Policy Analysts	Sections 1.4, 1.5 → Section 2.5 (regulatory constraints) → Section 5.5 (business rules) → Section 6 (compliance)
Product/Business Stakeholders	Sections 1.4, 2.2, 2.3 → Section 4.x.1 subsections only (skip REQ tables) → Section 5.5

Readers unfamiliar with Nepal-specific operating context should read Section 2.5 and Appendix A (Glossary) before Section 4, as caregiver-verification and payment-routing logic assume that context.

1.4 Product Scope

Nepal's remittance economy is structurally tied to labour migration: a very large share of working-age Nepalis are employed in the Gulf Cooperation Council states, Malaysia, South Korea (under the EPS scheme), Japan, Australia, and increasingly the United States and Europe. This has produced a demographic pattern common across many labour-sending districts: households where the primary income earner is abroad and the remaining household members are elderly parents, often without any adult child physically present in Nepal for months or years at a stretch.

Sahayatri's corporate goal is to convert a portion of the existing remittance flow — currently spent informally on ad hoc, unverifiable arrangements with neighbours, distant relatives, or unlicensed local attendants — into a formal, auditable, insured care-services market. The product's societal aim is threefold:

1. **Verified physical presence.** Give diaspora family members cryptographically-timestamped, GPS-tagged, photographic proof that a real, background-checked caregiver visited their parent, on the schedule they paid for.

2. **Emergency response continuity.** Reduce the time between an elderly recipient's medical or safety emergency and the arrival of help, in a country where formal ambulance dispatch coverage is inconsistent outside Kathmandu Valley, Pokhara, and a handful of other urban centers.
3. **Formalized, taxable local income for caregivers.** Create a structured gig/agency income channel for Nepali caregivers (many of whom are themselves nurses, ANMs — Auxiliary Nurse Midwives — or trained home-health aides) that pays predictably in NPR while being billed to sponsors abroad in hard currency.

Quantitative RC-1.0 targets (12 months post-launch, per Product's business case):

- 5,000 active paying diaspora subscriber households.
- 1,200 verified, background-checked caregivers onboarded across the top 15 labour-sending districts (Jhapa, Morang, Sunsari, Dhanusha, Siraha, Rupandehi, Kailali, Kanchanpur, Kaski, Chitwan, Kathmandu, Lalitpur, Bhaktapur, Makwanpur, Sindhupalchok).
- Median SOS acknowledgment time under 90 seconds; median dispatch-to-scene time under 25 minutes in Kathmandu Valley catchment, under 60 minutes in other district headquarters.
- 98.5% of scheduled visits producing valid, geo-fenced proof-of-visit records within the scheduled window (accounting for legitimate rural connectivity delays absorbed by offline sync).
- Caregiver payout reliability of 99.9% of scheduled batch payout runs completing without manual intervention.

Explicitly **out of scope for RC-1.0**: telemedicine/doctor video consultation marketplace (reserved for v1.2), integration with Nepal's national HMIS (Health Management Information System), and wearable IoT vitals ingestion beyond manually-logged readings (see Appendix C).

1.5 References

- IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications* (structural template basis for this document).
- IEEE Std 29148-2018, *Systems and software engineering — Life cycle processes — Requirements engineering*.
- **Senior Citizens Act, 2063 (2006) and Senior Citizens Regulations, 2065**, Government of Nepal — governing rights, welfare entitlements, and care obligations toward elderly citizens; referenced for caregiver conduct standards and mandatory elder-abuse reporting obligations.
- **Public Health Service Act, 2075 (2018)**, Government of Nepal — referenced for caregiver clinical-scope-of-practice boundaries (i.e., what a non-licensed home caregiver may vs. may not do, such as medication administration limits).

- **Nepal Rastra Bank (NRB) Foreign Exchange (Regulation) Act, 2019** and associated NRB directives on inward remittance and foreign currency service billing to residents/non-residents — referenced in Section 2.5 for cross-border billing constraints.
 - **Individual Privacy Act, 2075 (2018)**, Government of Nepal — referenced in Section 6 for data retention and consent requirements.
 - **Electronic Transactions Act, 2063 (2006)** — referenced for digital signature/audit-log admissibility of proof-of-visit records.
 - Stripe API Reference (<https://docs.stripe.com/api>) and PayPal Developer Docs — for diaspora billing integration (Section 3.3).
 - eSewa Merchant API v2 and Khalti Payment Gateway API documentation — for local caregiver payout rails (Section 3.3).
 - PostGIS 3.4 Manual — spatial indexing, `ST_DWithin`, geo-fence polygon operations (Section 2.4, 4.3, 4.4).
 - Twilio Programmable SMS/Voice API Docs, Agora.io RTC SDK Docs, Google Maps Platform (Distance Matrix, Geocoding, Places) API Docs.
 - OWASP Mobile Application Security Verification Standard (MASVS) v2.0 — baseline for Section 5.3 security requirements.
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2. Overall Description

2.1 Product Perspective

Sahayatri is a new, greenfield, three-sided marketplace platform. It is not a modification of an existing legacy system. It consists of:

Client tier:

- Three React Native mobile applications sharing a common component/design-token library but with role-specific navigation trees and offline data models (WatermelonDB per app, synchronized against the backend via a pull/push delta-sync protocol, see Section 2.5 and 4.3).

Application/API tier:

- A Node.js (Express) monolith-with-modular-boundaries REST API for RC-1.0 (explicitly *not* microservices at launch — see 2.5 constraint on team size and operational maturity), exposing versioned REST endpoints under `/v1/`.
- A dedicated **Socket.io cluster** (backed by a Redis adapter for cross-node pub/sub) handling all real-time bidirectional channels: SOS alert broadcast, live caregiver GPS location streaming during an active visit, live chat/video-call signaling handoff to Agora.io, and push-notification fan-out coordination.

- **Background worker fleet** (BullMQ on Redis) responsible for: scheduled visit reminder generation, offline-sync conflict resolution jobs, payout batch runs (eSewa/Khalti disbursement), currency FX-rate refresh jobs, subscription renewal/dunning jobs, and SOS escalation-timeout watchdogs.

Data tier:

- **PostgreSQL 16** with the **PostGIS 3.4** extension as the system of record. PostGIS is used for: geo-fenced proof-of-visit validation (was the caregiver's check-in GPS point actually within an acceptable radius of the CareRecipient's registered home point, given Nepal's lack of formal street addressing — see 2.5), nearest-emergency-facility lookups, and caregiver-to-recipient distance/ETA precomputation.
- **Redis** for session/cache, BullMQ queue storage, and Socket.io pub/sub adapter.
- **AWS S3 / Cloudinary** for immutable, append-only storage of proof-of-visit photographs and video-call recordings (SOS-triggered calls only, for dispute resolution), with signed-URL time-boxed access.

External gateway tier (all accessed via dedicated internal adapter modules, never called directly from business logic, to allow provider swap without core rewrites):

- **Stripe and PayPal** — diaspora-side subscription billing in USD/AUD/GBP/EUR (and other supported settlement currencies as the diaspora base expands).
- **eSewa and Khalti** — NPR-denominated caregiver payout and any domestic Nepal-based family co-payer top-ups.
- **Twilio** — international SMS/voice fallback for SOS notification when push notification and WebSocket delivery both fail (critical redundancy path, see 4.4).
- **Agora.io** — HD video/voice calling for family-to-recipient calls and SOS-triggered emergency video bridge to Emergency Coordinators.
- **Google Maps Platform** — geocoding of caregiver-entered addresses (since formal postal addressing is unreliable — see 2.5), Distance Matrix for ETA calculation, Places API for nearest hospital/health-post lookup fallback when internal PostGIS facility dataset is incomplete for a given district.

A high-level data-flow is: Diaspora Family App → REST API → Postgres (Subscription/VisitSchedule) → BullMQ (visit reminder job) → Push/SMS → Caregiver App → REST API (check-in with GPS+photo) → S3 (photo) + Postgres (VisitLog, PostGIS point) → Socket.io (broadcast "visit completed" to subscribed Family App) → Family App (dashboard update). The SOS flow is described in full in Section 4.4.

2.2 Product Functions

At a high level, Sahayatri performs six operational functions:

1. **Identity & Trust Verification** — KYC for diaspora payers (basic, for payment fraud prevention), and a multi-stage background-check + document-verification + in-person/video interview pipeline for caregivers before they may accept any paid visit.
2. **Care Plan Subscription & Multi-Currency Billing** — recurring foreign-currency subscription management for family payers, automatic FX conversion and ledger reconciliation, and NPR-denominated caregiver payout runs, with an escrow-like holding period between visit completion and caregiver payout release.
3. **Visit Scheduling & Proof-of-Visit** — recurring and ad hoc visit scheduling, geo-fenced GPS check-in/check-out, mandatory timestamped photograph capture, and structured wellness-check questionnaires completed by the caregiver each visit.
4. **Emergency SOS & Dispatch Coordination** — multi-channel emergency trigger (in-app button, physical BLE pendant, caregiver-initiated escalation), automatic nearest-caregiver and nearest-facility resolution, human Emergency Coordinator hand-off, and full audit trail.
5. **Remote Family Dashboard & Health Telemetry** — longitudinal view for diaspora family of visit history, manually-logged vitals trends (blood pressure, blood glucose, weight), medication-adherence checklists, and caregiver notes/photos.
6. **Trust & Safety / Dispute Resolution** — rating/review system (bidirectional — families rate caregivers, caregivers can flag unsafe/abusive situations), missed-visit penalty and refund logic, and an admin-operated dispute queue.

2.3 User Classes and Characteristics

User Class	Technical Proficiency	Primary Device Context	Access Privileges	Key Needs
Diaspora Family Payer	Medium-High. Comfortable with smartphone apps and international payment methods, but often on shift work (e.g., 12-hr construction shifts in Qatar) with irregular connectivity windows.	Personal smartphone abroad; occasionally shared family device.	Own subscription(s), view/rate visits for their linked CareRecipient(s), initiate SOS, video-call recipient, manage billing, invite co-payers (e.g., siblings splitting cost).	Trust that money translates into real, verifiable care; fast emergency notification regardless of time-zone difference; simple foreign-currency billing without hidden FX loss.
Elderly Care Recipient	Low, often first-generation smartphone users. May have vision/hearing/dexterity limitations. Rural connectivity is frequently 2G/3G.	Low-cost Android device or the optional BLE SOS pendant with no screen at all.	Trigger SOS, receive/answer video calls, view large-font simplified schedule of upcoming visits (no billing or admin access).	Physical safety, dignity, minimal cognitive load, ability to summon help with a single large button even without literacy.
Caregiver / Nurse	Medium. Many are trained ANMs or CMAs (Community Medical Assistants) already familiar with basic mobile	Mid-range Android device; frequently operates in	View/accept assigned visits, GPS check-in/out, upload proof photos, log vitals and notes, trigger SOS	Reliable offline queuing so a visit isn't "lost" due to

User Class	Technical Proficiency	Primary Device Context	Access Privileges	Key Needs
	data-entry apps used by NGOs/health posts, but device quality and mobile data budgets vary widely.	low-bandwidth rural zones.	escalation on recipient's behalf, view own payout ledger.	no signal at the recipient's home; fast, predictable payout; clear scope-of-practice boundaries to avoid liability.
Verification Admin	High. Internal platform operations staff.	Desktop web admin console (out of scope for mobile client detail, but backend API-driven).	Review/approve/reject caregiver KYC and background-check submissions, suspend accounts, manage dispute queue.	Efficient document review queue, audit trail, ability to flag fraud patterns (e.g., duplicate national ID numbers, reused photos).
Emergency Coordinator	High. Internal 24/7 operations staff (shifts).	Desktop web console + mobile app parity for on-call staff.	Receive escalated SOS events, view live caregiver/recipient GPS, initiate three-way call bridge, dispatch nearest available caregiver or contact local emergency services/hospital, close out incident with report.	Sub-minute situational awareness, reliable multi-channel alerting (their own device must not be a single point of failure), clear

User Class	Technical Proficiency	Primary Device Context	Access Privileges	Key Needs
				escalation runbook UI.

2.4 Operating Environment

Mobile Client Runtime:

- React Native 0.74+, targeting **iOS 15.0+** and **Android 8.0 (API 26)+** — the Android floor is set low deliberately, since a large share of caregivers and elderly recipients in rural Nepal use older, budget Android handsets.
- Offline data layer: **WatermelonDB** (SQLite-backed) on-device, chosen over plain AsyncStorage/Realm for its built-in sync-adaptor primitives suited to the delta-sync protocol described in 2.5.
- Push notifications: **FCM** (Android) and **APNS** (iOS), with Twilio SMS as the tertiary fallback channel described in 4.4.

Backend Runtime:

- **Node.js 20 LTS**, Express 4.x, deployed as containerized services (Docker) behind a load balancer, minimum 3 API nodes for RC-1.0 to satisfy the availability target in 5.4.
- **Socket.io 4.x** cluster with the Redis adapter (`socket.io-redis`) for multi-node broadcast consistency; sticky-session-free design so any node can serve any client's reconnect.
- **PostgreSQL 16** with **PostGIS 3.4**, provisioned as a managed instance with at least one hot standby replica in a separate availability zone, plus point-in-time recovery (PITR) with a minimum 14-day retention window.
- **Redis 7.x** for cache, session, BullMQ queues, and Socket.io adapter (run as a separate cluster from the queue Redis to avoid noisy-neighbor eviction of session data by queue traffic — a P1 operational separation for RC-1.0).
- **BullMQ** workers deployed as separate horizontally-scaled worker containers (not co-located with API request-handling containers), to isolate payout-batch and SOS-watchdog job processing from user-facing request latency.

Network Resilience: Given that a large share of caregiver and elderly-recipient client traffic originates on **2G/3G/congested rural 4G** networks, the operating environment explicitly assumes: median mobile RTT to the nearest edge/API region of 400-1200ms, packet loss up to 5% as a normal (not exceptional) condition, and multi-hour connectivity gaps as a routine (not edge-case) condition that the offline-sync design (Section 2.5, 4.3) must absorb gracefully rather than merely "retry."

2.5 Design and Implementation Constraints

- **REQ-COMP-001 [P0]** — All foreign-currency billing to diaspora payers **MUST** be structured as payment for a "digital platform subscription service" rather than a direct domestic remittance instrument, to avoid classification under Nepal Rastra Bank's remittance-channel licensing regime (which Sahayatri, as a tech platform rather than a licensed remittance company, does not hold). The Stripe/PayPal merchant-of-record entity **MUST** be a foreign (non-Nepal) incorporated entity; a separate Nepal-incorporated subsidiary **MUST** hold the eSewa/Khalti merchant accounts used solely for domestic NPR disbursement to caregivers, with an inter-company services agreement documenting the fund flow for NRB compliance review. This bifurcated-entity constraint directly shapes the ledger schema in Section 6: every `Subscription` (foreign-currency) and every `CaregiverPayout` (NPR) **MUST** be modeled as economically linked but legally and technically separate ledger entries, never a single cross-currency transaction row.
- **REQ-COMP-002 [P0]** — Caregiver onboarding **MUST** collect and verify a Nepali citizenship certificate or national ID number, and **MUST** cross-check against Nepal's criminal record disclosure process (via a manual document-review step performed by a Verification Admin, since no real-time national criminal-record API exists for automated checking as of RC-1.0). This constrains the verification workflow (4.1) to include an inherently manual, non-automatable admin review stage with SLA tracking rather than an instant automated approval.
- **REQ-DATA-001 [P0]** — Because **Nepal lacks a standardized, geocodable street-addressing system** in the majority of municipalities and rural wards (addresses are commonly expressed as ward number + landmark description, e.g., "Ward 4, near Shree Saraswati school, Kavre"), the `CareRecipient` location **MUST** be stored primarily as a PostGIS `POINT` geometry captured via **in-person GPS pinning during onboarding** (the caregiver or family member physically stands at the home and drops a pin in-app), with the free-text landmark description stored as a secondary, human-readable field only — never as the primary geocoding input. Google Maps Geocoding API **MAY** be used to reverse-geocode the point for display purposes only, and its forward-geocoding (address-to-point) **MUST NOT** be trusted as authoritative for geo-fence validation.
- **REQ-DATA-002 [P0]** — The geo-fence radius used to validate a caregiver's check-in as "at the recipient's home" **MUST** be configurable per-recipient (default 150 meters) rather than a single global constant, because GPS accuracy in dense urban Kathmandu apartment blocks versus open rural terrain varies significantly, and because some recipients live in multi-story buildings where horizontal GPS accuracy alone is insufficient (see REQ-FUNC-304 for the manual-override-with-photo-evidence fallback).

- **REQ-OFFLINE-001 [P0]** — All three mobile clients **MUST** function fully for core visit/SOS workflows with zero network connectivity for a minimum of 72 hours, queuing all writes locally (WatermelonDB) and syncing via delta-sync on reconnect. A caregiver completing a full visit (check-in, questionnaire, photo, check-out) entirely offline **MUST** produce a locally-timestamped record indistinguishable in business logic from an online-created one, with the only difference being a `synced_at` timestamp that trails the `created_at` timestamp — the system **MUST NOT** reject or discount a late-synced but validly-signed offline visit record.
- **REQ-PERF-001 [P1]** — Given median rural bandwidth constraints, proof-of-visit photo uploads **MUST** be client-side compressed (target: under 400KB per photo at acceptable diagnostic/verification quality) before upload, and the mobile client **MUST** support upload resumption after connectivity interruption rather than restart-from-zero.
- **REQ-FUNC-battery** (*cross-referenced under 4.3*) — Continuous GPS tracking during an active visit **MUST** use adaptive-frequency location sampling (e.g., every 45-90 seconds rather than continuous streaming) to constrain battery drain on caregiver devices, since caregivers may complete 4-6 visits per day on a single charge without reliable access to charging infrastructure between visits.
- **REQ-COMP-003 [P1]** — All caregiver conduct standards, mandatory elder-abuse/neglect reporting triggers, and the platform's minimum-visit-frequency defaults **MUST** align with obligations described in the **Senior Citizens Act, 2063** and its 2065 Regulations, including the Act's provisions on the duty of care and reporting of neglect; the Verification Admin console **MUST** provide a dedicated abuse/neglect-report escalation path distinct from the standard dispute queue (see 4.1, 5.5).
- **REQ-COMP-004 [P1]** — Caregivers who are not licensed nurses/ANMs **MUST** be restricted by the app's UI and business rules from logging or being assigned tasks that constitute administering prescription medication beyond reminding/observing self-administration, consistent with scope-of-practice boundaries implied by the **Public Health Service Act, 2075**; the `Caregiver` schema **MUST** carry a `licensure_tier` enum (`UNLICENSED_ATTENDANT`, `CMA`, `ANM`, `STAFF_NURSE`) gating which `VisitTask` types are assignable.

2.6 User Documentation

RC-1.0 ships with: (1) an in-app, skippable onboarding walkthrough per client app (Family, Care, Elder Companion), localized in **English and Nepali (Devanagari script)**; (2) a Caregiver Training Manual (PDF, distributed during the in-person verification interview stage of 4.1) covering proof-of-visit protocol, SOS escalation procedure, and elder-abuse reporting obligations; (3) a versioned public REST API specification (OpenAPI 3.1) for internal engineering and any future authorized third-party integrators; (4) a Family FAQ/Help Center covering billing, refund policy, and how proof-of-visit verification works, localized in English, Nepali, and — given the GCC diaspora concentration — an Arabic-

script quick-reference card for the small subset of family payers more comfortable in Arabic than English (translation of key billing terms only, not full app localization in RC-1.0).

2.7 Assumptions and Dependencies

- It is assumed that **Stripe and PayPal maintain support for the target diaspora settlement currencies** (USD, AUD, GBP, EUR, and CAD) at launch; any additional corridor (e.g., Korean Won for the large EPS-scheme worker population in South Korea) is a dependency on Stripe's local payment-method/currency roadmap and may require a fallback wire-transfer-linked manual reconciliation flow if unsupported at RC-1.0.
 - It is assumed **eSewa and Khalti maintain sufficient API uptime and payout-batch throughput** to disburse caregiver payments on the committed schedule (Section 5.5); a dependency-risk mitigation is that the payout worker **MUST** support both providers with automatic failover routing per caregiver's registered preferred wallet.
 - It is assumed that **caregiver-owned Android devices have functioning GPS hardware**; devices failing a GPS-accuracy self-test during onboarding are flagged and the caregiver is required to remediate (e.g., enable high-accuracy location mode) before activation.
 - **Emergency dispatch is a coordination and information-relay function, not a direct ambulance-operating function.** Sahayatri does not own or operate ambulances; it depends on existing local hospital/health-post/ambulance-service phone-dispatch infrastructure, which varies significantly in response capability outside Kathmandu Valley. This dependency is a known and disclosed platform limitation, not a solvable engineering constraint, and is reflected in differentiated SLA targets (1.4) between Kathmandu Valley and other districts.
 - It is assumed Twilio maintains SMS/voice delivery capability across all target diaspora countries' carrier networks and within Nepal's domestic carrier networks (Ncell, NTC) for the SOS fallback path.
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3. External Interface Requirements

3.1 User Interfaces

- **REQ-UI-001 [P0]** — All three client apps **MUST** support both light and dark theme, with theme following system preference by default and manual override available in settings.
- **REQ-UI-002 [P0]** — The Elder Companion app **MUST** implement a "high-contrast, large-touch-target" mode as the default (not opt-in) rendering mode: minimum 18pt base font size, minimum 64x64dp touch targets, a maximum of one primary action per screen, and a persistent, always-visible, non-scrolling SOS button rendered as a fixed bottom-anchored element on every screen of the app.

- **REQ-UI-003 [P0]** — All user-facing text across all three apps **MUST** be fully localized in **English and Nepali (Devanagari script, नेपाली)**, with language selectable independently per device (a family member's phone may run in English while the elderly recipient's device runs in Nepali) and stored as a per-user, not per-device, preference synced from the backend.
- **REQ-UI-004 [P1]** — The Family Dashboard **MUST** provide an interactive map view (Google Maps Platform SDK) showing the CareRecipient's registered home pin, the assigned caregiver's live location during an active visit (with explicit "location sharing active" indicator visible to the caregiver on their own screen, satisfying transparency/consent expectations), and pins for the nearest 3 emergency-capable facilities as precomputed by the backend (Section 4.4).
- **REQ-UI-005 [P1]** — Caregiver check-in/out screens **MUST** display a real-time visual geo-fence indicator (e.g., a radius circle on a mini-map with the caregiver's current position dot) so the caregiver receives immediate feedback if they are outside the acceptable check-in radius, rather than only receiving a rejection message after submission.

3.2 Hardware Interfaces

- **REQ-INT-101 [P0]** — Camera interface: the Caregiver app **MUST** invoke the native camera (not gallery picker) for proof-of-visit photo capture, embedding EXIF GPS coordinates and a server-issued visit-session nonce (rendered as a small on-photo visible watermark plus embedded metadata) to deter after-the-fact photo substitution.
- **REQ-INT-102 [P0]** — GPS/GNSS receiver interface: all location-dependent flows **MUST** request high-accuracy location mode and **MUST** reject (with a clear, non-technical error message) any location reading with a reported accuracy radius greater than 50 meters for check-in validation purposes, prompting the user to move to open sky or wait for a better fix, with a manual-override-with-photo-evidence escalation path (REQ-FUNC-304) for genuinely GPS-denied environments (e.g., deep valley terrain, thick-walled stone housing).
- **REQ-INT-103 [P1]** — Optional BLE SOS Pendant: the Elder Companion app **MUST** support pairing with a simple single-button BLE peripheral (no display, replaceable coin-cell battery, IP54 water resistance minimum) that, on press, wakes the paired phone's Sahayatri app in the background and triggers the same SOS flow as the in-app button (Section 4.4), including when the phone screen is locked.
- **REQ-INT-104 [P0]** — Local SQLite cache storage (via WatermelonDB) **MUST** be used as the interface between the app's UI layer and all locally-queued, not-yet-synced data, per REQ-OFFLINE-001.

3.3 Software Interfaces

System	Provider	Purpose	Integration Notes
Diaspora Billing	Stripe (primary), PayPal (secondary)	Recurring subscription charges in USD/AUD/GBP/EUR; supports cards, and region-relevant local payment methods surfaced via Stripe's payment-method APIs.	Webhooks (<code>invoice.paid</code> , <code>invoice.payment_failed</code> , <code>customer.subscription.deleted</code>) drive the internal <code>Subscription</code> state machine (Section 4.2). Idempotency keys required on all charge-creation calls to prevent double-billing on client retry.
Local Payout	eSewa Merchant API v2 (primary), Khalti Payment Gateway API (secondary/failover)	NPR disbursement to caregiver wallets on a batch payout schedule (Section 5.5).	Payout batch jobs MUST reconcile provider callback/webhook status before marking a <code>CaregiverPayout</code> row as <code>SETTLED</code> ; a <code>FAILED</code> payout MUST automatically retry via the secondary provider before falling to manual admin review.
SMS/Voice Fallback	Twilio Programmable SMS & Voice	Tertiary SOS notification channel when push+WebSocket delivery is unacknowledged within the timeout window (Section 4.4); also used for OTP-based phone verification at signup.	International SMS delivery rates and carrier filtering vary by destination country; the system MUST track per-message delivery-receipt status and escalate to voice call if SMS delivery is not confirmed within 60 seconds during an active SOS.
Video/Voice Calling	Agora.io RTC SDK	Family-to-recipient video calls; SOS-triggered three-way bridge between recipient, nearest caregiver, and	Token-based channel auth issued by the backend per call session, scoped to a single channel name and short TTL, never a long-lived credential shipped to the client.

System	Provider	Purpose	Integration Notes
		Emergency Coordinator.	
Media Storage	AWS S3 (primary) with Cloudinary as an image-transformation/CDN layer in front of S3	Immutable storage of proof-of-visit photos and SOS-call recordings.	S3 objects for proof-of-visit photos MUST be written with Object Lock (governance mode) for the minimum retention period defined in Section 6, to preserve dispute-resolution evidentiary integrity.
Location/Mapping	Google Maps Platform (Geocoding, Distance Matrix, Places, Maps SDK)	Reverse-geocoding for display, ETA/distance calculation for nearest-caregiver and nearest-facility resolution, interactive map rendering.	See REQ-DATA-001 — forward geocoding is display-only, never authoritative for geo-fence math, which is computed server-side via PostGIS against the recipient's pinned point.

3.4 Communications Interfaces

- **REQ-INT-201 [P0]** — All client-server REST communication **MUST** use HTTPS with **TLS 1.3** (TLS 1.2 accepted only as a fallback for older Android WebView components, never below 1.2), with HSTS enforced on all backend domains.
- **REQ-INT-202 [P0]** — All Socket.io real-time connections **MUST** use secure WebSockets (**WSS**), authenticated via a short-lived JWT passed during the Socket.io handshake (not as a persistent query-string token logged in access logs).
- **REQ-INT-203 [P0]** — Push notification delivery **MUST** use **FCM** (Android) and **APNS** (iOS) as primary channels, with payload size kept under provider limits and all SOS-class push notifications marked high-priority/time-sensitive to bypass Android battery-optimization deferral where the OS permits.
- **REQ-INT-204 [P1]** — The REST API **MUST** be versioned via URL path prefix (`/v1/...`), with a documented deprecation policy of minimum 6 months' notice before retiring a version, given the offline-first mobile clients may run outdated app versions for extended periods without forced update in low-connectivity regions.

4. System Features

4.1 Caregiver Verification & Onboarding Engine

4.1.1 Description & Priority

Priority: P0. This feature governs the multi-stage pipeline by which a caregiver applicant becomes an "Active, Bookable" caregiver on the platform. Because Sahayatri's core value proposition is *trust* — a family member abroad is paying a stranger to have physical access to their elderly parent's home — this pipeline is deliberately conservative, multi-stage, and includes mandatory human review steps that cannot be fully automated (per REQ-COMP-002). No caregiver may be assigned a visit, nor appear in family-facing search/browse results, until they reach the ACTIVE state.

4.1.2 Stimulus/Response Sequences

#	Stimulus	System Response
1	Applicant submits phone number	System sends OTP via Twilio SMS; applicant enters OTP; system creates Caregiver record in PENDING_DOCUMENTS state.
2	Applicant uploads citizenship/national ID document photo + selfie	System runs automated liveness/face-match check (basic client-side liveness capture + backend face-match confidence score); record moves to PENDING_ADMIN_REVIEW regardless of automated score (automated check is a fraud-signal input, not an approval gate).
3	Verification Admin opens applicant in review queue	System displays document images, face-match score, applicant-declared licensure tier and any uploaded nursing/ANM certification, and a checklist for manual criminal-background disclosure review.
4	Admin approves document stage	Record moves to PENDING_INTERVIEW ; system schedules/prompts an in-person or video interview slot.
5	Admin marks interview passed	Record moves to PENDING_TRAINING ; system unlocks the in-app Caregiver Training Manual acknowledgment module.
6	Applicant completes training acknowledgment quiz (minimum passing score)	Record moves to ACTIVE ; caregiver becomes bookable and appears in district-scoped family search results.
7	At any stage, admin identifies fraud signal (duplicate ID number, reused selfie hash across applicants, mismatched name/ID)	Record moves to REJECTED with a reason code; a rejected national ID number is flagged platform-wide against future re-application under a different phone number.
8	Post-activation, three or more substantiated family complaints or one elder-abuse/neglect report is logged against a caregiver	Record automatically moves to SUSPENDED pending admin investigation; all future assignments blocked immediately; existing scheduled visits are reassigned.

4.1.3 Functional Requirements

- **REQ-FUNC-001 [P0]** — The system **MUST** model caregiver onboarding as an explicit finite-state machine with states **PENDING_DOCUMENTS → PENDING_ADMIN_REVIEW → PENDING_INTERVIEW → PENDING_TRAINING → ACTIVE**, plus terminal/exception states

`REJECTED` and `SUSPENDED`), with all state transitions logged immutably (actor, timestamp, reason code) in a `CaregiverStatusHistory` table.

- **REQ-FUNC-002 [P0]** — The system **MUST** require, at minimum, one government-issued photo ID (citizenship certificate or national ID card) and one live selfie captured within the app (not gallery-uploaded) before a caregiver can leave `PENDING_DOCUMENTS`.
- **REQ-FUNC-003 [P0]** — The system **MUST** compute and store a perceptual hash of every submitted selfie and ID photo, and **MUST** flag to the admin review queue any new submission whose hash matches (within a similarity threshold) a photo already associated with a different applicant's national ID number, as a duplicate-identity fraud signal.
- **REQ-FUNC-004 [P0]** — A caregiver **MUST NOT** be assignable to any visit, nor visible in any family-facing search/browse endpoint, unless `Caregiver.status = 'ACTIVE'`.
- **REQ-FUNC-005 [P1]** — The system **MUST** capture a self-declared `licensure_tier` (per REQ-COMP-004) with optional certificate document upload for `CMA`, `ANM`, or `STAFF_NURSE` tiers; Verification Admins **MUST** manually confirm certificate authenticity before the declared tier is honored in `VisitTask` assignment gating — an unconfirmed declared tier defaults to `UNLICENSED_ATTENDANT` privileges.
- **REQ-FUNC-006 [P0]** — The system **MUST** provide a dedicated, differently-styled/escalation-prioritized admin queue entry point for elder-abuse/neglect reports (distinct from general service-quality disputes), per REQ-COMP-003, and **MUST** auto-suspend the reported caregiver's bookability pending investigation upon the first such report (not requiring three occurrences, unlike general complaints), since the risk asymmetry of a false-negative here is unacceptable.
- **REQ-FUNC-007 [P1]** — The system **MUST** track interview scheduling and completion via an `InterviewRecord` with a configurable SLA (default: 7 days from document approval to interview completion), and **MUST** surface overdue interview records in an admin operational dashboard, since delayed manual steps are the primary onboarding-funnel bottleneck given REQ-COMP-002's manual-review constraint.
- **REQ-FUNC-008 [P1]** — The Caregiver Training Manual acknowledgment module **MUST** require a passing score (configurable, default 80%) on a comprehension quiz covering proof-of-visit protocol, SOS escalation steps, and mandatory elder-abuse reporting obligations, before the record can transition to `ACTIVE`; failed attempts **MUST** be retrievable after a cool-down period (default 24 hours) with a randomized question subset to discourage rote memorization.
- **REQ-FUNC-009 [P2]** — *(Reserved for v1.1)* Integration with a future automated national criminal-record disclosure API, should Nepal's government make one

available; schema **MUST** reserve a `criminal_record_check_reference` nullable field on `Caregiver` for forward compatibility.

4.2 Multi-Currency Diaspora Subscription & Billing Engine

4.2.1 Description & Priority

Priority: P0. This feature manages the recurring, foreign-currency care-plan subscription purchased by one or more Diaspora Family Payers on behalf of a single CareRecipient, the currency-conversion and internal ledger reconciliation between the foreign-currency subscription revenue and the NPR-denominated caregiver payout obligation it funds, and the escrow-like holding mechanism that withholds caregiver payout release until a visit is confirmed complete.

4.2.2 Stimulus/Response Sequences

#	Stimulus	System Response
1	Family payer selects a care plan tier (e.g., "3 visits/week + unlimited SOS") and enters payment method	System creates a Stripe (or PayPal) subscription object, creates a local <code>Subscription</code> record in <code>PENDING_FIRST_PAYMENT</code> state, and awaits webhook confirmation.
2	Stripe webhook <code>invoice.paid</code> received	<code>Subscription</code> moves to <code>ACTIVE</code> ; system computes the funded visit-credit allotment for the billing period and begins allowing visit scheduling against it.
3	A scheduled visit completes and is confirmed (proof-of-visit validated per 4.3)	System creates a <code>CaregiverPayoutLineItem</code> in <code>PENDING_PAYOUT</code> state at the caregiver's standard district-tier NPR rate, holding it for the configured escrow window (default: 24 hours) to allow a family dispute window before payout batching.
4	Escrow window elapses with no dispute flagged	<code>CaregiverPayoutLineItem</code> becomes eligible for the next scheduled payout batch run (Section 5.5).
5	Stripe webhook <code>invoice.payment_failed</code> received	System places <code>Subscription</code> into <code>PAST_DUE</code> , triggers a dunning notification sequence to the payer (and any co-payers), and — if unresolved after the configured grace period (default 5 days) — suspends future visit scheduling (already-scheduled visits within the current paid period still execute) and moves <code>Subscription</code> to <code>SUSPENDED_NONPAYMENT</code> .
6	Family payer invites a co-payer (e.g., sibling) to split subscription cost	System sends invitation; upon acceptance, subscription cost is split per configured percentage across linked <code>Payer</code> records, each billed separately by their own configured payment method for their share.
7	Family payer cancels subscription	System schedules cancellation effective at end of current paid period (no mid-period pro-rated refund by default, per REQ-FUNC-215 business rule in 5.5), cancels future visit scheduling beyond that date, and preserves historical visit records indefinitely for the family's later reference/export.

4.2.3 Functional Requirements

- **REQ-FUNC-201 [P0]** — The system **MUST** model `Subscription` as a state machine: `PENDING_FIRST_PAYMENT → ACTIVE ⇌ PAST_DUE → (SUSPENDED_NONPAYMENT | ACTIVE)`, plus `CANCELLING → CANCELLED`, with all transitions webhook- or admin-action-driven and logged.
- **REQ-FUNC-202 [P0]** — The system **MUST** support billing in at minimum USD, AUD, GBP, and EUR at RC-1.0 launch, with the settlement currency selected at subscription creation and immutable thereafter (a payer wishing to change currency **MUST** cancel and re-subscribe, to avoid FX-timing ambiguity in historical reporting).
- **REQ-FUNC-203 [P0]** — Every `Subscription` **MUST** be linked to exactly one `CareRecipient` and **MAY** be linked to multiple `Payer` accounts (co-payers) with a configurable cost-split percentage per payer summing to 100%.
- **REQ-FUNC-204 [P0]** — The system **MUST** never store or transmit raw payment card data; all card handling **MUST** occur via Stripe.js/Stripe Elements (or PayPal's equivalent client SDK) tokenization, with only the resulting payment-method token stored server-side, in line with PCI-DSS SAQ-A scope minimization (see 5.3).
- **REQ-FUNC-205 [P0]** — Every confirmed, non-disputed completed visit **MUST** generate exactly one `CaregiverPayoutLineItem` denominated in NPR at the caregiver's currently-effective district-tier rate card (rate cards versioned by effective date, never mutated retroactively for already-generated line items).
- **REQ-FUNC-206 [P0]** — The system **MUST** enforce the escrow hold window (default 24 hours, configurable per district based on observed dispute-latency patterns) between visit confirmation and payout-batch eligibility, during which a family payer **MAY** flag the visit as disputed (e.g., "caregiver did not actually visit," "photo appears reused"), which **MUST** freeze that specific line item pending admin dispute resolution (Section 5.5) without blocking other unrelated line items for the same caregiver from proceeding to payout.
- **REQ-FUNC-207 [P0]** — The system **MUST** maintain the foreign-currency `Subscription`/`Invoice` ledger and the NPR `CaregiverPayoutLineItem`/`CaregiverPayout` ledger as **separate, non-commingled ledger tables**, per REQ-COMP-001, linked only via a non-transactional `funding_reference` foreign key for reporting/reconciliation purposes — no single database transaction **MUST** ever attempt to atomically debit a foreign-currency ledger row and credit an NPR ledger row, since they are legally distinct corporate entities' books.
- **REQ-FUNC-208 [P1]** — Dunning notifications on payment failure **MUST** be sent to all linked payers (not just the primary), via both push notification and email, with escalating urgency framing at day 1, day 3, and day 5 of the grace period, and **MUST** clearly state the consequence (visit-scheduling suspension) rather than only requesting updated payment info.

- **REQ-FUNC-209 [P1]** — The system **MUST** compute and display, at subscription checkout, an all-in-cost breakdown including the platform's FX conversion reference and any card-network foreign-transaction implications known to be typical for the payer's card-issuing region, to satisfy transparency expectations for a cost-sensitive diaspora user base — this is an informational disclosure, not a guarantee of the payer's actual bank-charged FX rate.
- **REQ-FUNC-210 [P1]** — Co-payer invitation links **MUST** expire after 7 days if unaccepted, and a subscription with an unaccepted co-payer invitation **MUST** continue billing the inviting payer at 100% until acceptance, never leaving a funding gap.

4.3 GPS-Timestamped & Photo Proof Visit Management System

4.3.1 Description & Priority

Priority: P0. This is the platform's core trust mechanism: converting a scheduled care visit into a cryptographically-timestamped, geo-validated, photographically-evidenced record that a diaspora family member — who cannot personally verify anything happening 6,000+ kilometers away — can trust.

4.3.2 Stimulus/Response Sequences

#	Stimulus	System Response
1	Scheduled visit reminder time reached	Background worker pushes reminder notification to assigned caregiver 30 minutes before the scheduled window opens.
2	Caregiver taps "Check In" at the recipient's home	App captures GPS coordinates + accuracy radius; if online, submits to server for real-time geo-fence validation against <code>CareRecipient.home_point</code> (PostGIS <code>ST_DWithin</code>); if offline, stores the attempt locally with <code>pending_sync</code> flag and proceeds optimistically, deferring hard validation to sync time.
3	Geo-fence validation passes	Server (or, if offline, the local app pending later server confirmation) opens an active <code>VisitLog</code> in <code>IN_PROGRESS</code> state; caregiver app unlocks the visit checklist (wellness questionnaire, vitals entry, photo capture requirement).
4	Geo-fence validation fails (caregiver outside configured radius)	App displays the real-time geo-fence indicator (REQ-UI-005) and blocks check-in submission with a clear message; if the caregiver believes the failure is a GPS-accuracy artifact (e.g., inside thick-walled housing), they may invoke the manual-override-with-photo-evidence path (REQ-FUNC-304), which requires an additional landmark photo and is flagged for lightweight async admin spot-review rather than blocking the visit.
5	Caregiver completes wellness questionnaire, logs any manually-measured vitals, and captures the required proof-of-visit photo	App locally validates all required fields are present, then either submits immediately (online) or queues (offline).
6	Caregiver taps "Check Out"	App captures a second GPS reading (must also pass geo-fence, or same override path); <code>VisitLog</code> moves to <code>COMPLETED</code> ; visit duration is computed from check-in to check-out timestamps.
7	Device reconnects to network after an offline visit	Delta-sync protocol uploads all queued records in creation order; server re-validates geo-fence server-side authoritatively (the earlier offline "optimistic pass" is not trusted blindly) and reconciles <code>synced_at</code> vs <code>created_at</code> ; a server-side geo-fence failure on a synced offline record is routed to the same manual-override review queue rather than silently rejected, since the caregiver has no opportunity to retry an already-departed visit.

#	Stimulus	System Response
8	Visit successfully syncs and confirms	Socket.io broadcasts a <code>visit.completed</code> event to any online, subscribed Family App client; push notification sent regardless of Family App's online status.

4.3.3 Functional Requirements

- **REQ-FUNC-301 [P0]** — Every `VisitLog` **MUST** record: caregiver ID, care-recipient ID, scheduled window, actual check-in timestamp + GPS point + accuracy radius, actual check-out timestamp + GPS point + accuracy radius, geo-fence validation result (pass / manual-override / server-corrected-fail), wellness questionnaire responses, any logged vitals, and at least one proof-of-visit photo reference.
- **REQ-FUNC-302 [P0]** — The system **MUST** enforce, server-side, that no `VisitLog` can be marked `COMPLETED` without at least one associated proof-of-visit photo that passed the EXIF-GPS-and-nonce-watermark check (REQ-INT-101); a visit missing this **MUST** remain in an `INCOMPLETE` state visible to the family as such, never silently upgraded to appear as a normal completed visit.
- **REQ-FUNC-303 [P0]** — All timestamps **MUST** be captured and stored in UTC with the device's local timezone offset recorded alongside for display purposes, given caregivers, recipients, and family payers routinely span 3–6 different timezones simultaneously (Nepal, GCC states, Malaysia, Korea, Australia).
- **REQ-FUNC-304 [P1]** — The system **MUST** provide a "manual override" check-in path, usable only when automated geo-fence validation fails, requiring: (a) a supplementary photo showing a recognizable exterior landmark of the home, (b) a caregiver-entered short text justification, and (c) automatic flagging into a lightweight async Verification Admin spot-review queue (not blocking visit completion, but subject to retroactive investigation and, per REQ-FUNC-006-style asymmetric risk handling, immediate escalation if a pattern of repeated overrides for the same caregiver emerges).
- **REQ-FUNC-305 [P0]** — The offline delta-sync protocol **MUST** be idempotent: re-submission of an already-synced local record (e.g., due to a retried request after an ambiguous network timeout) **MUST NOT** create a duplicate `VisitLog`, enforced via a client-generated UUID-based idempotency key created at record-creation time (not sync time), persisted locally so retries always reuse the same key.
- **REQ-FUNC-306 [P1]** — Proof-of-visit photos **MUST** be retained with immutable storage (Object Lock governance mode per Section 3.3) for the minimum dispute-resolution retention window defined in Section 6, and **MUST NOT** be deletable by any client-side action, including caregiver or family account deletion requests, until that retention window lapses.

- **REQ-FUNC-307 [P1]** — The wellness questionnaire **MUST** be configurable per care-recipient (e.g., a diabetic recipient's family may enable a blood-glucose logging field; a mobility-limited recipient's family may enable a fall-risk checklist item) rather than a single fixed global form, managed via a `QuestionnaireTemplate` join to `CareRecipient`.
- **REQ-FUNC-308 [P1]** — Continuous location sampling during an `IN_PROGRESS` visit (for the family's live-map view, REQ-UI-004) **MUST** use the adaptive-frequency sampling described in Section 2.5 (battery constraint) and **MUST** cease immediately upon check-out, with no location tracking of the caregiver outside the bounds of an active, recipient-linked visit session (a deliberate privacy boundary — caregivers are not tracked between visits).

4.4 Real-time Emergency SOS & Dispatch Subsystem

4.4.1 Description & Priority

Priority: P0. This is the platform's highest-stakes feature. It must function reliably under exactly the adverse network conditions (rural low-bandwidth, elderly users with limited digital literacy, potential caregiver unavailability) that are the platform's normal operating baseline, not an edge case.

4.4.2 Stimulus/Response Sequences

#	Stimulus	System Response
1	Elder Companion in-app SOS button pressed, or paired BLE pendant button pressed	App immediately (a) attempts to open a WebSocket SOS event to the backend with the recipient's current GPS point, (b) in parallel, attempts a push notification wake regardless of WebSocket success, (c) starts a local on-device countdown/acknowledgment-wait UI so the recipient sees something is happening even with no connectivity yet.
2	Backend receives SOS event	System creates an <code>EmergencyAlert</code> record in <code>TRIGGERED</code> state, immediately fans out via Socket.io to: (a) all linked family payers' apps, (b) the currently-assigned/nearest available caregiver (computed via PostGIS nearest-neighbor query against active caregivers' last-known visit-session location or registered home-base location), and (c) the on-duty Emergency Coordinator queue.
3	No acknowledgment (by any of family, caregiver, or coordinator) within 45 seconds	System escalates: triggers Twilio SMS to all linked family payers' registered phone numbers, and triggers a Twilio voice call to the on-duty Emergency Coordinator's registered phone (bypassing app-level notification entirely) as a redundant device-independent channel.
4	An Emergency Coordinator acknowledges and opens the incident	<code>EmergencyAlert</code> moves to <code>ACKNOWLEDGED</code> ; coordinator's console displays recipient's live GPS, nearest 3 precomputed emergency facilities with Distance-Matrix-derived ETA, nearest available caregiver's current distance/ETA, and one-tap actions to initiate an Agora.io three-way call bridge (recipient + nearest caregiver + coordinator) or to place an outbound call to a local hospital/health-post contact.
5	Coordinator dispatches nearest available caregiver	System notifies that caregiver with priority override (bypassing their normal visit-acceptance flow) and tracks their live location en route; <code>EmergencyAlert</code> moves to <code>DISPATCHED</code> .
6	Situation resolved (caregiver/coordinator confirms recipient is safe, or hospital transfer confirmed)	Coordinator closes the incident with a structured resolution report (outcome category, actions taken, any hospital/facility involved); <code>EmergencyAlert</code> moves to <code>RESOLVED</code> , immutably logged with full timeline for later review and, where applicable, for Senior Citizens Act neglect/abuse-reporting compliance documentation.

#	Stimulus	System Response
7	Recipient or caregiver triggers SOS while device has zero connectivity	Local app queues the SOS trigger as maximum-priority and retries transmission continuously (more aggressively than normal offline-sync backoff) on any connectivity blip, including a brief cell-tower handoff; if the device has any SMS capability even without data connectivity, the app additionally attempts a raw SMS trigger to a dedicated short-code/number as a data-connectivity-independent fallback.

4.4.3 Functional Requirements

- **REQ-FUNC-401 [P0]** — The system **MUST** model `EmergencyAlert` as a state machine: `TRIGGERED → ACKNOWLEDGED → DISPATCHED → RESOLVED`, with a parallel `ESCALATED` flag (not a replacing state) settable at any point once the 45-second no-acknowledgment timeout fires, and full immutable timeline logging of every state transition and notification-channel attempt with delivery-confirmation status.
- **REQ-FUNC-402 [P0]** — SOS trigger sources (in-app button, BLE pendant, caregiver-initiated escalation on behalf of a recipient) **MUST** all funnel into the identical `EmergencyAlert` creation and fan-out logic — no source-specific shortcut path that skips coordinator visibility.
- **REQ-FUNC-403 [P0]** — The 45-second no-acknowledgment escalation timeout (REQ-FUNC-401 stimulus #3) **MUST** be implemented as a durable, crash-safe watchdog job (BullMQ delayed job, not an in-memory timer), since the backend node handling the initial event **MUST NOT** be a single point of failure for the escalation guarantee.
- **REQ-FUNC-404 [P0]** — The system **MUST** maintain an always-current, precomputed nearest-caregiver lookup per active/on-duty caregiver's last-known location (updated at each visit check-in/out and, during an active visit, at each adaptive-frequency location ping per REQ-FUNC-308) via a PostGIS spatial index, such that nearest-caregiver resolution during an SOS event completes in well under 1 second even with several thousand concurrently tracked caregivers.
- **REQ-FUNC-405 [P0]** — SOS-class push notifications **MUST** be sent with maximum provider priority (FCM high-priority data message; APNS with `interruption-level: critical`) where the app holds the appropriate entitlement) to maximize likelihood of delivery even under OS-level battery/notification throttling.
- **REQ-FUNC-406 [P0]** — The Twilio SMS/voice fallback (stimulus #3) **MUST** trigger automatically and **MUST NOT** require any human platform-operator action to initiate — the automated escalation is itself the redundancy; a human coordinator's job is response coordination, not triggering the redundant alert.

- **REQ-FUNC-407 [P1]** — The system **MUST** maintain a dataset of known emergency-capable facilities (hospitals, primary health-care centers, health posts with ambulance access) per district, sourced initially via manual data-collection/partnership onboarding, with Google Places API used only as a supplementary fallback lookup for districts with incomplete internal data coverage — internal curated data **MUST** take precedence when both exist, since Places API listings are not verified for actual emergency-response capability.
- **REQ-FUNC-408 [P1]** — The Emergency Coordinator console **MUST** support simultaneous handling of multiple concurrent `EmergencyAlert` incidents with clear visual prioritization (e.g., unacknowledged alerts approaching their escalation timeout surfaced above already-`DISPATCHED` incidents), since a single coordinator may be on duty covering multiple districts simultaneously in RC-1.0's staffing model.
- **REQ-FUNC-409 [P1]** — Raw-SMS-triggered SOS (stimulus #7, data-connectivity-independent fallback) **MUST** be parseable by a dedicated inbound-SMS webhook that maps the sending phone number to a known `CareRecipient` or `Caregiver` account and creates an `EmergencyAlert` with a `source: 'sms_fallback'` tag, with the understanding that this path carries no GPS coordinate (SMS alone cannot carry precise location) and therefore **MUST** immediately surface the recipient's last-known registered home point as a location estimate to the coordinator, explicitly labeled as "last known, not live."

4.5 Remote Family Dashboard & Health Telemetry Tracking

4.5.1 Description & Priority

Priority: P1. This feature is the ongoing, day-to-day retention driver for diaspora family payers between emergencies: a longitudinal, trustworthy view of their parent's well-being, visit history, and health trends, designed for a user checking in during scarce off-shift moments across a significant timezone gap.

4.5.2 Stimulus/Response Sequences

#	Stimulus	System Response
1	Family payer opens dashboard	App displays: next scheduled visit countdown, most recent completed visit summary (photo, wellness snapshot, caregiver note), and a trend chart of any enabled vitals (blood pressure, glucose, weight) over the selected time range.
2	Family payer taps into a specific past <code>VisitLog</code>	App displays full detail: timestamps, geo-fence validation status, proof photo, full questionnaire responses, caregiver's free-text notes, and a rating/review prompt if not already rated.
3	Family payer submits a 1-5 star rating and optional text review for a completed visit or caregiver	System stores the review, updates the caregiver's rolling average rating (visible in future family-facing caregiver search/browse), and — if rating is 2 stars or below — automatically routes the review into the dispute/quality-review admin queue for follow-up, distinct from the elder-abuse escalation path (which requires an explicit abuse-flag, not merely a low rating).
4	Family payer sets a medication-adherence checklist item (e.g., "morning blood pressure medication")	Item appears on the caregiver's per-visit checklist for future visits; caregiver marks observed/reminded status each visit (not administers, per REQ-COMP-004 scope-of-practice constraint — the caregiver reminds/observes, doesn't administer, unless licensure_tier permits and the family/care plan explicitly enables it).
5	Family payer initiates a video call to the recipient	System establishes an Agora.io channel, sends a high-priority notification/ring to the Elder Companion app (and, if configured, simultaneously to the currently-assigned caregiver's device if a visit is in progress, so the caregiver can help the recipient answer).

4.5.3 Functional Requirements

- **REQ-FUNC-501 [P1]** — The dashboard **MUST** render a trend chart for each enabled vitals type over selectable ranges (7/30/90 days), sourced from manually-entered `VitalsReading` records logged by caregivers during visits, clearly labeled as "manually logged by caregiver during scheduled visits" rather than implying continuous medical-grade monitoring, to avoid over-promising clinical accuracy.
- **REQ-FUNC-502 [P1]** — The system **MUST** compute and surface a simple visit-adherence percentage (visits completed on schedule vs. visits scheduled) over the current billing period, as a core trust/transparency metric on the dashboard home screen.
- **REQ-FUNC-503 [P1]** — Caregiver rolling average ratings **MUST** be computed from a minimum sample size (default: 5 ratings) before being displayed publicly in search/browse results, to avoid a single early negative review disproportionately

affecting a newly-active caregiver's visibility; ratings below the minimum sample threshold **MUST** display as "New Caregiver" rather than a potentially misleading average.

- **REQ-FUNC-504 [P1]** — A review of 2 stars or below **MUST** automatically create a `QualityReviewCase` visible in the admin dispute queue within 5 minutes of submission, separate from and with lower default urgency than the elder-abuse escalation path (REQ-FUNC-006), and **MUST** notify the assigned caregiver's supervising admin contact (if the caregiver is agency-affiliated rather than independent) for coaching follow-up.
 - **REQ-FUNC-505 [P2]** — *(Reserved for v1.1)* Family-configurable custom checklist items beyond the default questionnaire template set; RC-1.0 ships with admin-managed template presets only, not full family self-service template authoring.
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5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **REQ-PERF-101 [P0]** — REST API p95 response latency **MUST** be under 500ms for standard read endpoints and under 1200ms for write endpoints involving PostGIS geo-fence computation, measured at the load balancer, excluding client-side network transit time (which, per Section 2.4, is assumed to routinely add 400-1200ms on top for rural clients — the backend budget is deliberately tight to leave headroom for that unavoidable transit cost).
- **REQ-PERF-102 [P0]** — Socket.io SOS event broadcast (server receipt to all subscribed client delivery attempts dispatched) **MUST** complete server-side fan-out in under 200ms for up to 50 concurrently subscribed clients per `EmergencyAlert` (family members, caregiver, coordinators combined).
- **REQ-PERF-103 [P1]** — Database query benchmarks: the nearest-caregiver PostGIS spatial query (REQ-FUNC-404) **MUST** return results in under 150ms at a scale of 10,000 actively tracked caregiver location points; the nearest-facility lookup **MUST** return in under 150ms at a scale of 5,000 curated facility records.
- **REQ-PERF-104 [P1]** — Offline queue synchronization: upon reconnection after an offline period, the delta-sync protocol **MUST** complete upload of up to 50 queued records (typical multi-day caregiver backlog) within 30 seconds on a throttled 3G-equivalent connection (384kbps), leveraging the client-side photo pre-compression (REQ-PERF-001) to keep payload size manageable.
- **REQ-PERF-105 [P1]** — Payout batch runs (Section 5.5) **MUST** process up to 5,000 `CaregiverPayoutLineItem` disbursements within a 2-hour scheduled batch window, with automatic partial-failure isolation (a single failed disbursement **MUST NOT** block or roll back unrelated successful disbursements in the same batch).

5.2 Safety Requirements

- **REQ-SEC-safety-01 [P0]** — Failover routine for an unacknowledged SOS: if no family member, caregiver, or coordinator acknowledges an **EmergencyAlert** within the 45-second timeout **and** the subsequent Twilio SMS/voice escalation also receives no delivery confirmation within an additional 90 seconds, the system **MUST** automatically escalate to a secondary, always-staffed "critical incident" coordinator phone tree (a documented internal on-call rotation, technically implemented as an ordered list of phone numbers dialed sequentially by the voice-escalation worker until one is answered), ensuring no single unavailable individual can silently absorb a failed emergency response.
- **REQ-SEC-safety-02 [P0]** — The system **MUST NOT** allow a caregiver's account suspension (e.g., due to a fraud or abuse investigation) to silently cancel an already-**IN_PROGRESS** visit without immediate reassignment prompting for the family and, where the recipient is mid-visit with a now-suspended caregiver, immediate Emergency Coordinator notification for a welfare check — suspension of trust in a caregiver **MUST** trigger a safety-check workflow, not merely an administrative account-state change.
- **REQ-SEC-safety-03 [P1]** — Physical safety protocol: caregivers **MUST** complete training-module content (REQ-FUNC-008) covering basic fall-response, choking-response, and "when to call local emergency services directly versus escalating in-app first" guidance, since in a genuine life-threatening emergency the app's escalation chain **MUST NOT** be positioned as a substitute for the caregiver directly calling local emergency/ambulance services when seconds matter — the in-app SOS system is a coordination and family-notification layer, not a replacement for direct emergency calling, and training materials **MUST** make this priority order explicit.

5.3 Security Requirements

- **REQ-SEC-001 [P0]** — All data at rest in PostgreSQL and S3 **MUST** be encrypted using **AES-256**; database-level encryption **MUST** be enabled at the storage-volume level at minimum, with field-level encryption additionally applied to national ID numbers, raw phone numbers, and any stored payment-method tokens.
- **REQ-SEC-002 [P0]** — Authentication **MUST** use **JWT access tokens with a short TTL (15 minutes)** paired with **refresh token rotation**: each refresh-token use issues a new refresh token and invalidates the prior one, with reuse-detection (an already-used, invalidated refresh token presented again) triggering immediate full session revocation across all that account's devices, as an anti-token-theft measure.
- **REQ-SEC-003 [P0]** — Payment handling **MUST** remain within **PCI-DSS SAQ-A scope** by never having Sahayatri's own servers touch raw card data (REQ-FUNC-204); any expansion of payment functionality that would touch raw card data **MUST** trigger a full PCI-DSS scope re-assessment before implementation.

- **REQ-SEC-004 [P0]** — The system **MUST** implement **Role-Based Access Control (RBAC)** with the following minimum role/permission matrix:

Role	View Own Visits	View Assigned Visits	Trigger SOS	Approve Caregiver KYC	View All Users' PII	Issue Refunds	Access Payout Ledger
Diaspora Family Payer	✓ (linked recipient only)	✗	✓	✗	✗ (own linked recipient only)	✗ (request only)	✗
Elderly Care Recipient	✓ (own, simplified view)	✗	✓	✗	✗	✗	✗
Caregiver	✗	✓ (own assigned only)	✓ (on behalf of recipient)	✗	✗	✗	✓ (own payout history only)
Verification Admin	✗	✗	✗	✓	✓ (scoped to verification-relevant fields)	✗	✗
Emergency Coordinator	✗	✓ (during active incident only)	N/A (receives, doesn't trigger)	✗	✗ (incident-scoped only)	✗	✗
Finance/Ops Admin (implied support role)	✗	✗	✗	✗	✗	✓	✓

- **REQ-SEC-005 [P0]** — Every access to another user's PII by any internal Admin/Coordinator role **MUST** be logged in an immutable audit table (actor, target

record, timestamp, justification field for out-of-normal-workflow access), reviewable for compliance auditing.

- **REQ-SEC-006 [P1]** — Mobile client local storage (WatermelonDB) **MUST** be encrypted at rest on-device using platform-native secure storage primitives (iOS Keychain-backed encryption key, Android Keystore-backed encryption key) for the SQLite database file, since offline-queued records may sit on a caregiver's or family member's device for extended periods per REQ-OFFLINE-001.
- **REQ-SEC-007 [P1]** — The backend **MUST** implement rate limiting on all authentication and OTP-request endpoints, and **MUST** implement anomaly detection on SOS-trigger frequency per account (an account triggering an implausible number of SOS events in a short window is flagged for review, balancing abuse-prevention against never delaying or blocking a genuine trigger — the flag is post-hoc review only, never a pre-trigger block).

5.4 Software Quality Attributes

Attribute	Quantitative Target
Availability	99.9% monthly uptime for the core API and Socket.io SOS-path specifically (≈ 43 minutes/month maximum downtime); 99.5% for non-critical dashboard/reporting endpoints.
Reliability	Zero data loss for any successfully-locally-persisted offline record (REQ-OFFLINE-001) — a synced-but-server-rejected record must be surfaced for caregiver/admin remediation, never silently dropped.
Maintainability	New REST endpoint additions must not require changes to more than 2 existing modules on average (modular monolith boundary discipline); minimum 80% unit test coverage on all business-logic modules (state machines, payout calculation, geo-fence validation).
Usability	Elder Companion app: a first-time, non-technical elderly user must be able to successfully trigger a test-SOS within 30 seconds of first opening the app, validated via usability testing with representative users during beta.
Adaptability	Payment gateway adapters (Stripe/PayPal, eSewa/Khalti) must be swappable behind their internal interface contracts without touching core Subscription/Payout state-machine logic, validated by the RC-1.0 architecture explicitly supporting the documented secondary-provider failover (Section 3.3).

5.5 Business Rules

- **REQ-FUNC-215 [P0]** — Subscription cancellation **MUST NOT** trigger a pro-rated refund for the remainder of the current paid billing period by default; a family payer

retains full visit-scheduling access through the period they already paid for, then the subscription lapses. Full refunds **MUST** only be issued by Finance/Ops Admin discretion in cases of platform-caused failure (e.g., no caregiver could be assigned for the entire paid period in that district).

- **REQ-FUNC-216 [P0]** — A **missed visit** (scheduled visit window closes with no `VisitLog` reaching `COMPLETED` or `INCOMPLETE`-with-partial-evidence state, and no caregiver-initiated reschedule request) **MUST** trigger: (a) immediate family notification, (b) automatic pro-rated visit-credit restoration to the subscription's remaining period (the family effectively gets that visit "back" as an extra visit later in the period), and (c) a strike recorded against the responsible caregiver, with 3 strikes in a rolling 60-day window triggering automatic suspension pending admin review.
 - **REQ-FUNC-217 [P0]** — Caregiver payout batch runs **MUST** execute on a fixed weekly schedule (default: every Thursday, covering line items that cleared escrow by Wednesday end-of-day), with payout amount computed strictly from `CaregiverPayoutLineItem` records at the district-tier rate effective on the date the visit occurred — never the rate effective on the payout date, to avoid retroactively under- or over-paying caregivers due to mid-cycle rate-card updates.
 - **REQ-FUNC-218 [P1]** — The **emergency escalation hierarchy** for unacknowledged SOS events **MUST** follow the strict order: (1) assigned/nearest caregiver + all linked family payers simultaneously, (2) on-duty Emergency Coordinator queue, (3) automated Twilio SMS to family + automated Twilio voice call to on-duty coordinator, (4) secondary critical-incident coordinator phone tree (REQ-SEC-safety-01) — no step may be skipped even if an earlier step's channel is known to be degraded, since redundant simultaneous attempts (not sequential replacement) are the intended design.
 - **REQ-FUNC-219 [P1]** — District-tier caregiver rate cards **MUST** be reviewable and adjustable only by Finance/Ops Admin, versioned with an effective-date, and **MUST** provide a minimum 14-day advance-notice period to affected caregivers before a rate decrease takes effect (rate increases may take effect immediately), as a baseline fairness commitment given caregivers' income-planning dependency on predictable rates.
-

6. Other Requirements

6.1 Database Design Requirements & ERD Schema Outline

The core relational entities and their primary relationships are:

- `FamilyAccount` (1) —< `Payer` (N): a family account may have multiple linked payer users (co-payers/siblings).
- `Payer` (N) >—< (N) `Subscription`: many-to-many via a `SubscriptionPayer` join table carrying the cost-split percentage.

- **Subscription** (1) —< **CareRecipient** (1): one active subscription funds exactly one care recipient's care plan (a recipient may have subscription *history*, i.e., past cancelled subscriptions, but at most one **ACTIVE** subscription at a time).
- **CareRecipient** (1) —< **VisitSchedule** (N) —< **VisitLog** (N): a recurring schedule generates individual visit-log instances.
- **Caregiver** (1) —< **VisitLog** (N): a caregiver is assigned to and completes many visit logs over time.
- **VisitLog** (1) —< **ProofPhoto** (N), (1) —< **VitalsReading** (N), (1) — (1) **QuestionnaireResponse**.
- **VisitLog** (1) — (0..1) **CaregiverPayoutLineItem**: a completed, non-disputed visit generates exactly one payout line item.
- **CaregiverPayoutLineItem** (N) —< **CaregiverPayout** (1): many line items are aggregated into one batch payout transaction.
- **CareRecipient** (1) —< **EmergencyAlert** (N): a recipient may be the subject of many emergency alerts over time.
- **EmergencyAlert** (1) —< **EmergencyAlertTimeline** (N): immutable event-log entries per alert.
- **Caregiver** (1) —< **CaregiverStatusHistory** (N): immutable onboarding/status-transition audit trail.
- **CareRecipient** (1) — (1) **home_point** (PostGIS **GEOGRAPHY(POINT, 4326)**), per REQ-DATA-001.

All monetary-value columns **MUST** be stored as integer minor-units (e.g., cents, or NPR paisa) rather than floating-point, to avoid rounding-error accumulation across the payout and billing ledgers.

6.2 Data Retention Policies

- Proof-of-visit photos and **VisitLog** records: retained for a **minimum of 5 years** from visit date (dispute-resolution and, where relevant, potential legal/regulatory review under the Senior Citizens Act's elder-welfare provisions), stored under S3 Object Lock governance mode per REQ-FUNC-306.
- **EmergencyAlert** full timelines: retained **indefinitely** (never auto-deleted), given their potential relevance to ongoing welfare-monitoring and safety-pattern analysis for a given care recipient across their entire time on the platform.
- Payment/payout ledger records: retained per standard financial record-keeping practice, **minimum 7 years**, consistent with typical statutory bookkeeping retention expectations in both the diaspora-payer jurisdictions' consumer-protection norms and Nepal's domestic bookkeeping practice for the local subsidiary entity.
- Caregiver KYC documents (ID photos, selfies): retained for the duration of active platform status **plus 3 years** post-deactivation, then purged, balancing fraud-pattern-

detection utility against data-minimization principles under the **Individual Privacy Act, 2075**.

- Chat/call metadata (not full recordings, except SOS-triggered calls per Section 3.3): retained **1 year**, with SOS-triggered call recordings following the 5-year proof-of-visit-equivalent retention window given their dispute/safety-review relevance.

6.3 Compliance with Nepalese Privacy Laws

- All data-collection consent flows (caregiver onboarding, family payer signup, elderly recipient account creation — with consent captured from the recipient or their legally authorized family representative where the recipient's capacity is limited) **MUST** present clear, Nepali-language-available consent language describing what data is collected and why, satisfying the **Individual Privacy Act, 2075**'s consent and purpose-limitation principles.
 - Cross-border data transfer (diaspora payer PII processed by foreign-incorporated billing entity per REQ-COMP-001's bifurcated-entity structure) **MUST** be documented in a data-processing agreement between the foreign parent entity and the Nepal subsidiary, addressing the Privacy Act's expectations around data leaving domestic control where recipient data (the more sensitive, elder-welfare-relevant dataset) is concerned — the recipient's core welfare/location/health data **SHOULD** be architected to reside primarily in infrastructure under the Nepal subsidiary's operational control even where the broader application stack is centrally hosted, as a data-sovereignty-conscious design posture.
-

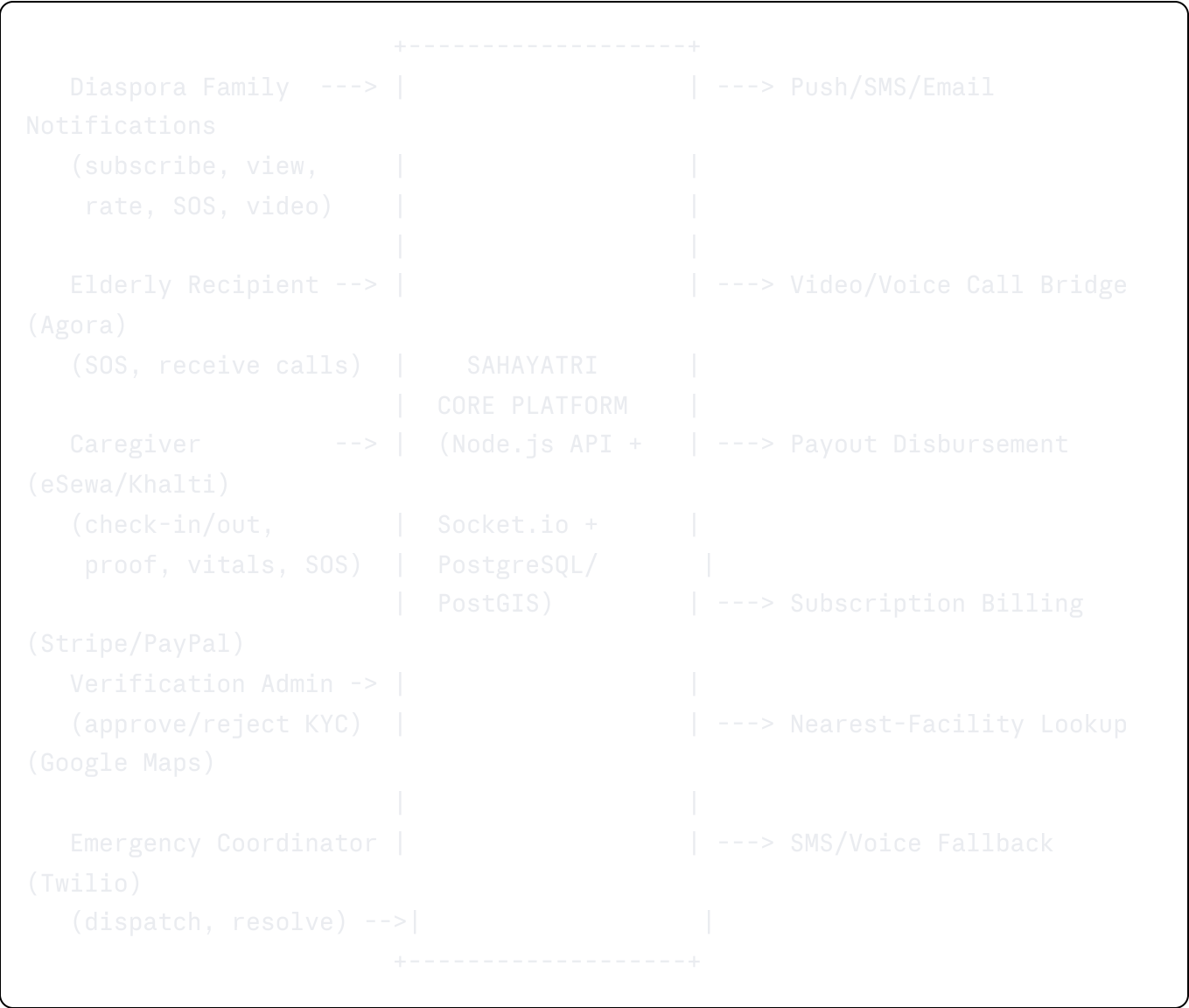
Appendix A: Glossary

Term	Definition
Diaspora	Nepali citizens residing and working outside Nepal, in this platform's context specifically the labour-migrant population in GCC states, Malaysia, South Korea, Japan, Australia, and other destination countries, who act as the platform's primary paying customer segment.
Proof-of-Visit	The combined evidentiary package (GPS-validated check-in/out, timestamped photograph, completed questionnaire) generated by a caregiver during a scheduled visit, serving as the platform's core trust artifact for the paying family.
PostGIS	A spatial-database extension for PostgreSQL enabling geographic object storage and spatial queries (distance, containment, nearest-neighbor), used here for geo-fence validation and nearest-caregiver/facility lookups.
WebSocket / Socket.io	A persistent, bidirectional network connection protocol (and, respectively, a Node.js library implementing it) used for real-time, low-latency event delivery such as SOS alerts and live location updates, as opposed to request/response REST polling.
Escrow (in this context)	The configurable holding period between a visit being marked complete and the corresponding caregiver payout line item becoming eligible for batch disbursement, during which a family dispute can freeze that specific payment.
KYC (Know Your Customer)	The identity-verification process applied at differing rigor levels to family payers (lightweight, fraud-prevention-oriented) and caregivers (extensive, safety-and-trust-oriented, including document and interview review).
Multi-tenant ledger	The architectural principle (Section 4.2, 6.1) of keeping the foreign-currency subscription ledger and the NPR payout ledger as separate, non-commingled sets of ledger tables reflecting two distinct legal entities, linked only by a non-transactional reference.
Geo-fence	A virtual, configurable-radius circular boundary around a CareRecipient 's pinned home GPS point, used to validate that a caregiver's check-in/check-out location is plausibly at the correct home.

Term	Definition
ANM (Auxiliary Nurse Midwife) / CMA (Community Medical Assistant)	Nepal-specific mid-level health-worker qualification tiers, referenced in the <code>licensure_tier</code> field gating which care tasks a caregiver may be assigned.
EPS Scheme	Employment Permit System — the bilateral labour-migration program under which a large population of Nepali workers are employed in South Korea, referenced as a target diaspora corridor.

Appendix B: Analysis Models

B.1 Data Flow Diagram — Level 0 (Context Diagram, text representation)



B.2 Data Flow Diagram — Level 1 (Proof-of-Visit Subsystem, text representation)

```
Caregiver App
|
| 1. Check-in (GPS + timestamp)
v
[Geo-Fence Validation Process] --uses--> PostGIS ST_DWithin against
CareRecipient.home_point
|
| 2a. Pass --> [Open VisitLog IN_PROGRESS]
| 2b. Fail --> [Manual Override Path] --flags--> Admin Spot-Review Queue
v
[Wellness Questionnaire + Photo Capture Process]
|
| 3. Check-out (GPS + timestamp)
v
[Close VisitLog COMPLETED] ---> [Generate CaregiverPayoutLineItem,
PENDING_PAYOUT]
|                                     |
| 4. Broadcast visit.completed       | 5. Escrow hold window (24h
default)
v                                     v
Family App (Socket.io + Push)      [Payout Batch Eligibility Queue] --->
eSewa/Khalti Payout Worker
```

B.3 Entity-Relationship Diagram (text representation)



B.4 Use Case Mapping (summary)

Use Case	Primary Actor	Related Requirements
Onboard as a verified caregiver	Caregiver, Verification Admin	4.1, REQ-FUNC-001-009, REQ-COMP-002
Subscribe to a care plan from abroad	Diaspora Family Payer	4.2, REQ-FUNC-201-210, REQ-COMP-001
Complete a scheduled visit with proof	Caregiver	4.3, REQ-FUNC-301-308, REQ-DATA-001/002
Trigger and resolve an emergency	Elderly Recipient, Caregiver, Emergency Coordinator	4.4, REQ-FUNC-401-409, REQ-SEC-safety-01-03
Monitor parent's well-being remotely	Diaspora Family Payer	4.5, REQ-FUNC-501-504
Receive weekly payout	Caregiver	5.5, REQ-FUNC-217-219
Review and resolve a service dispute	Verification Admin	4.1 (REQ-FUNC-006), 4.2 (REQ-FUNC-206), 4.5 (REQ-FUNC-504)

Appendix C: To Be Determined (TBD) List

The following are explicitly deferred beyond RC-1.0 and are noted here so their absence is a documented product decision rather than an oversight:

1. **Smart wearable IoT integration** (e.g., automatic fall-detection wearables, continuous heart-rate/SpO2 monitoring bands) — RC-1.0 supports only manually-logged vitals (REQ-FUNC-501); schema should reserve extensibility for a future `DeviceReading` ingestion table keyed to a `CareRecipient` with a `source_device_type` field.
2. **Automated AI health-anomaly detection** over logged vitals/questionnaire trends (e.g., flagging a concerning blood-pressure trend automatically to family/caregiver) — deferred pending sufficient logged-data volume and a defined clinical-oversight/liability model; RC-1.0's trend charts (REQ-FUNC-501) are descriptive/visual only, with no automated alerting layer.
3. **Automated national criminal-record disclosure API integration** for caregiver background checks (REQ-FUNC-009) — contingent on future government API availability; RC-1.0 relies entirely on manual admin document review (REQ-COMP-002).
4. **Telemedicine / doctor video-consultation marketplace** — explicitly out of scope per Section 1.4, reserved for v1.2 roadmap consideration.

5. **National HMIS (Health Management Information System) integration** — no current integration point exists; any future integration would require its own dedicated compliance and interoperability specification beyond this SRS's scope.
6. **Additional diaspora settlement currencies beyond USD/AUD/GBP/EUR at launch** (notably Korean Won for the EPS-scheme worker population, and various GCC-region local payment method nuances) — contingent on Stripe/PayPal roadmap support per the dependency noted in Section 2.7.
7. **Family self-service custom questionnaire/checklist authoring** (REQ-FUNC-505) — RC-1.0 ships admin-managed template presets only.
8. **Multi-tenant agency dashboards** for caregiver staffing agencies managing pools of their affiliated caregivers on the platform (referenced implicitly in REQ-FUNC-504's "supervising admin contact" concept) — RC-1.0 treats all caregivers as individually onboarded; formal agency-tenant tooling is a post-RC-1.0 consideration.